

Bowdoin Mathematics Teaching Demonstration

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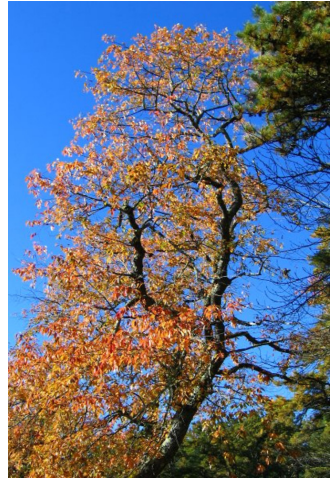
November 17, 2025



Scan for slides

Motivation: Black Cherry Trees

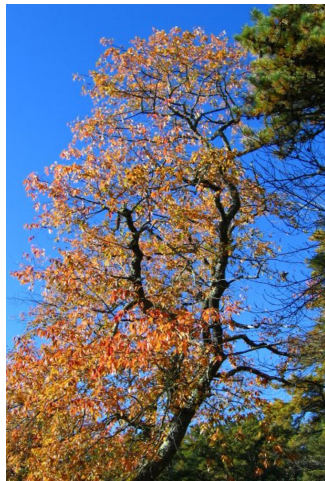
- Native North Carolina tree



source: nativeplanttrust.org

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- Interested in the size of these trees:
 - ▶ Tree height
 - ▶ Total tree volume

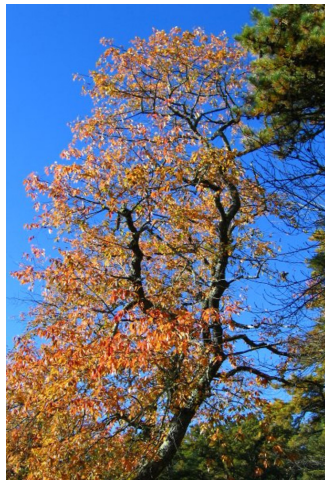


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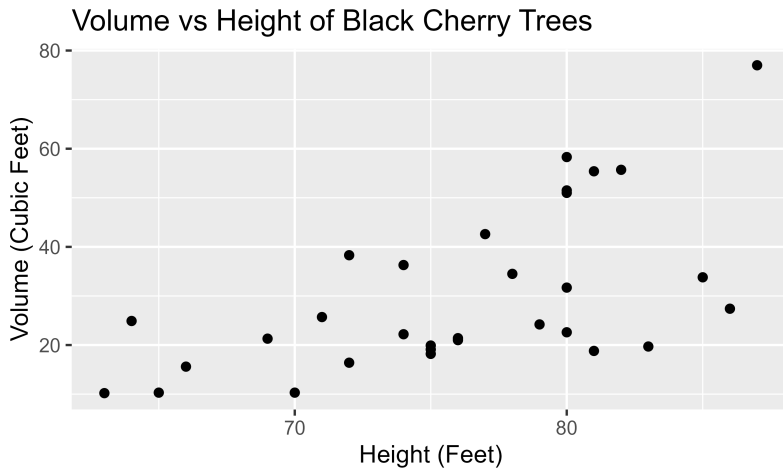
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Height (ft)	Volume (ft ³)
70	10.3
65	10.3
63	10.2
72	16.4
...	...



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Black Cherry Tree Data



Black Cherry Tree Questions

- We know how to analyze one variable at a time

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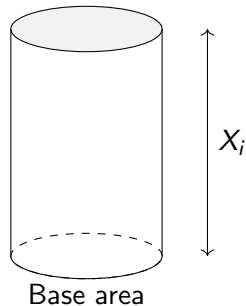
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 - ▶ Sample quantities: mean, median, mode, variance, interquartile range, etc
 - ▶ Inference about population quantities
- What if we want to analyze these two variables **together**?
 - ▶ What is the **association** between height and volume?
 - ▶ Given a tree's height, can we **predict** its volume?

Deterministic Relationship: Cylindrical Trees

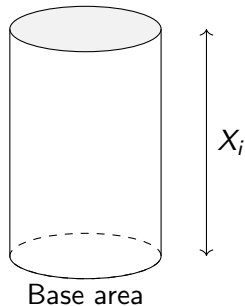
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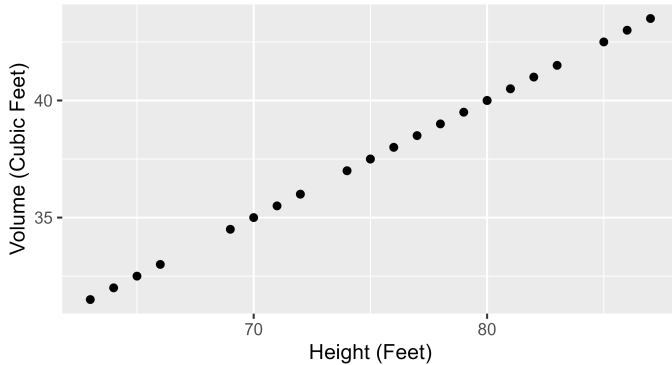
- Imagine that these trees have perfectly cylindrical trunks (with the same width), and that each tree has the same volume of branches
- Then we would have a **deterministic** relationship between volume and height:

$$\underbrace{Y_i}_{\text{volume of tree } i} = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} * \underbrace{X_i}_{\text{height of tree } i}$$



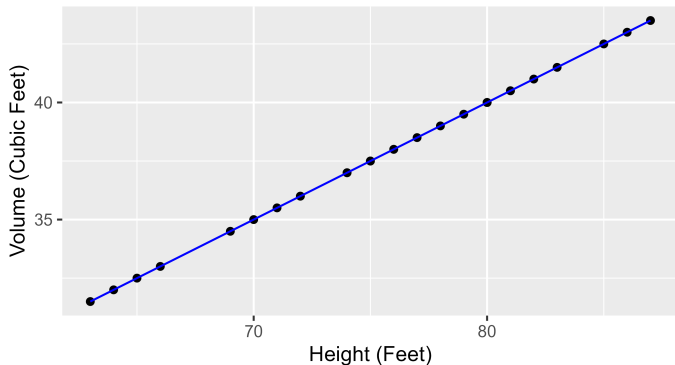
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Volume vs Height of Cylindrical Black Cherry Trees



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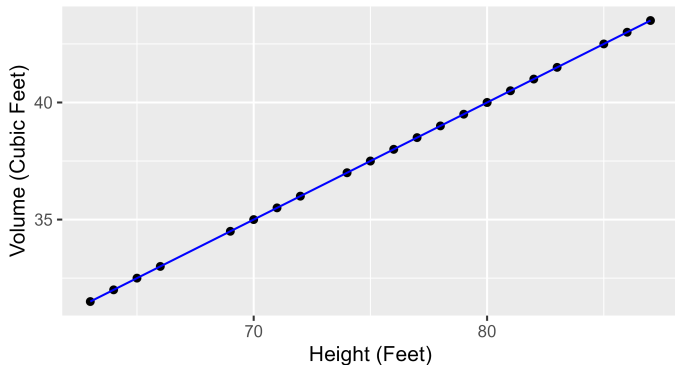
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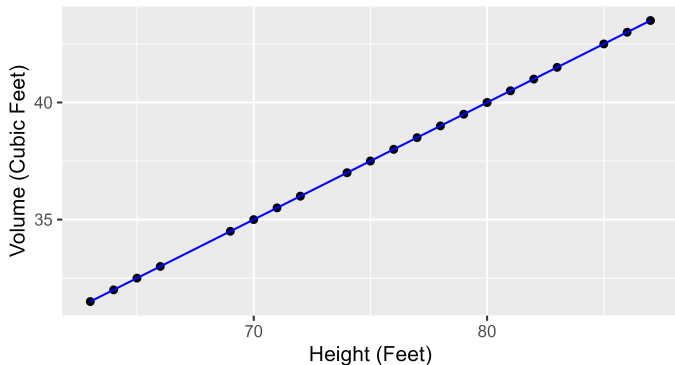


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More Realistic Trees

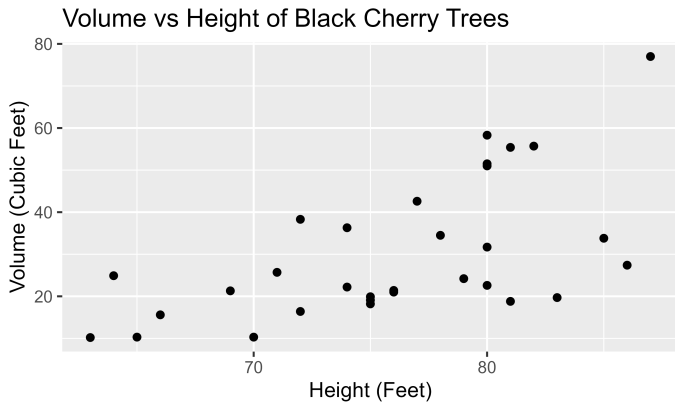
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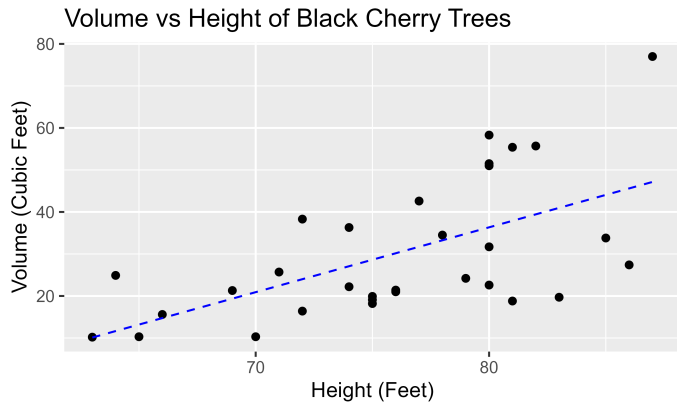
- In statistics, we almost never deal with **deterministic** relationships
- Instead, we consider **outcomes** to be **random**, but have **expectation** that depends on **covariates**

$$\underbrace{E(Y_i|X_i)}_{\text{expected volume of tree } i \text{ given its height}} = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} * \underbrace{X_i}_{\text{height of tree } i}$$

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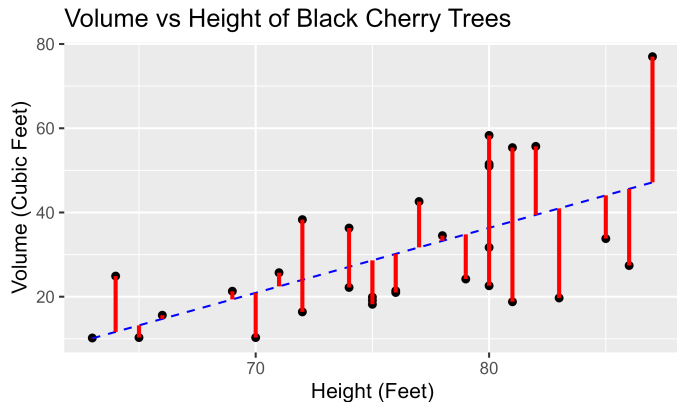
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$$\underbrace{Y_i}_{\text{volume of tree } i} = \underbrace{\beta_0 + \beta_1 X_i}_{\text{regression line}} + \underbrace{\epsilon_i}_{\text{random error}}$$

Linear Regression Model

Data: n pairs of (X_i, Y_i)

- Covariate X
- Outcome Y

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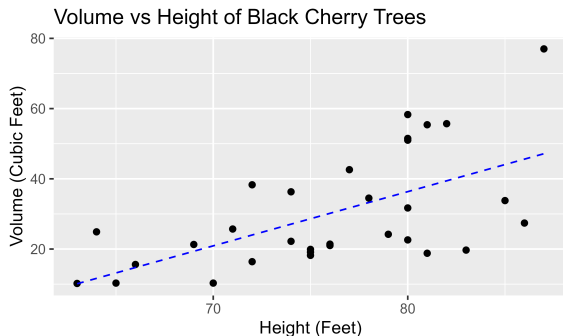
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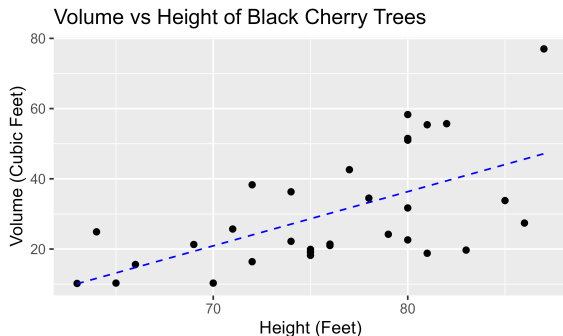
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- 5 The errors have a normal distribution:
 $\epsilon_i|X_i \sim N(0, \sigma^2)$

Interpreting Regression Model Parameters



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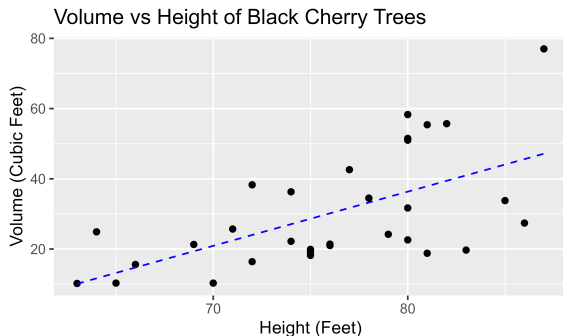
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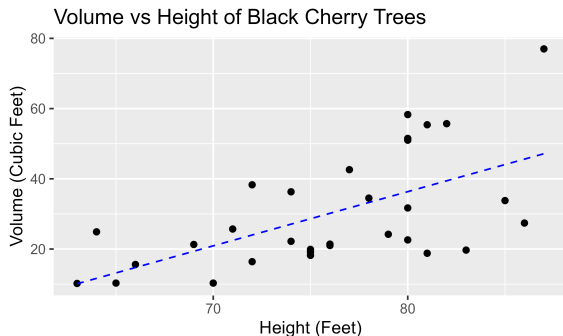
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What do these mean in the context of the Black Cherry trees?

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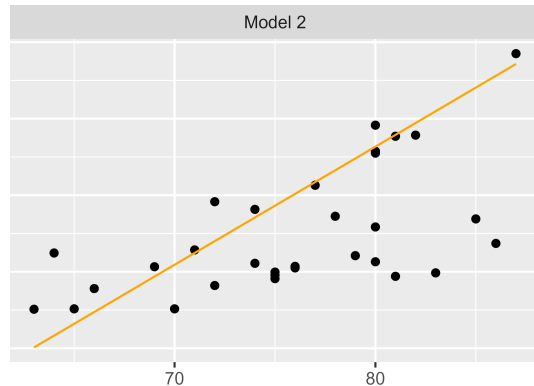
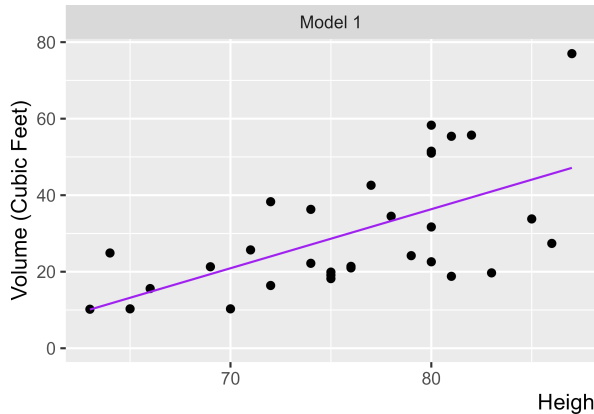
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 - ▶ **Least squares** estimation

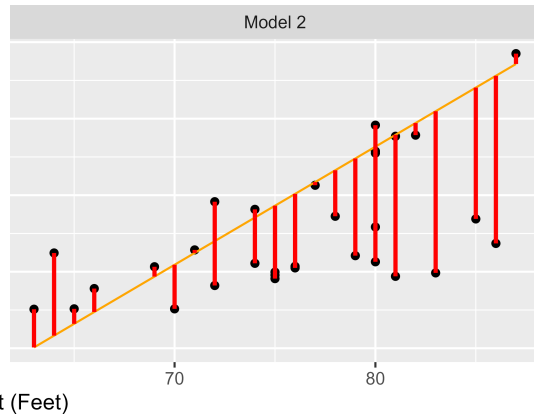
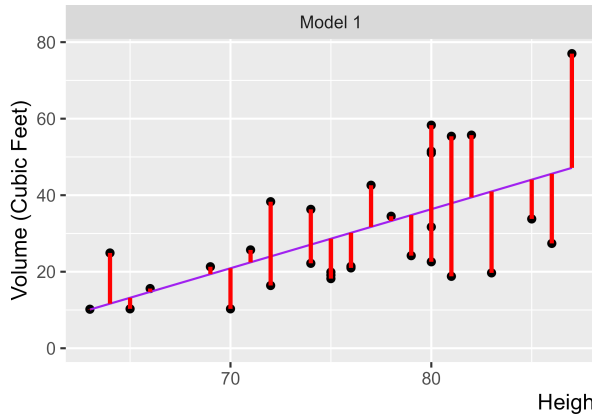
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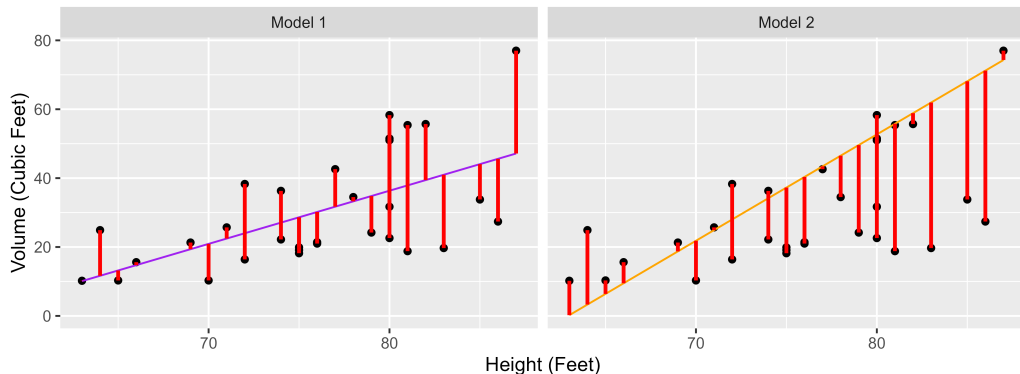
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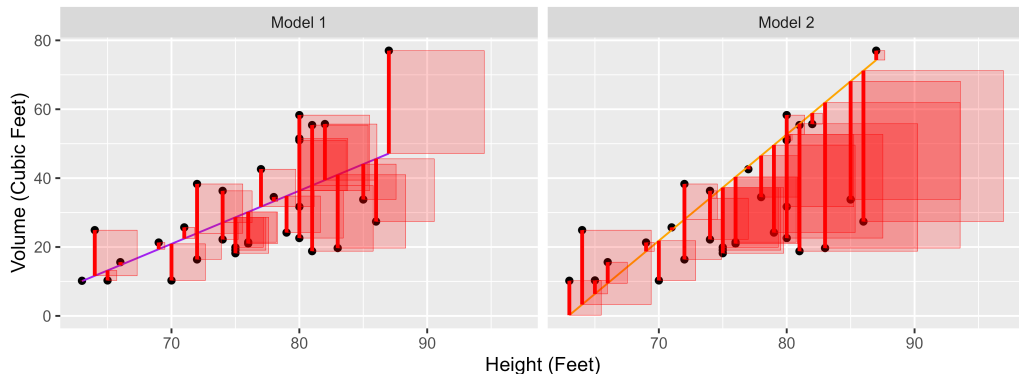
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- Use this to conduct **hypothesis tests** and find **confidence intervals** for β_0, β_1

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How would we interpret the results if $p = 0.01$? If $p = 0.1$?

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How would we interpret a confidence interval, e.g., $(-0.2, 0.8)$?

R Demo

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- What if we have more than one covariate?
- What if the outcome is binary?

References

- [1] Simon Sheather. *A Modern Approach to Regression with R*. Springer Texts in Statistics. Springer New York, New York, NY, 2009. ISBN 978-0-387-09607-0. doi: 10.1007/978-0-387-09608-7. URL <http://link.springer.com/10.1007/978-0-387-09608-7>.